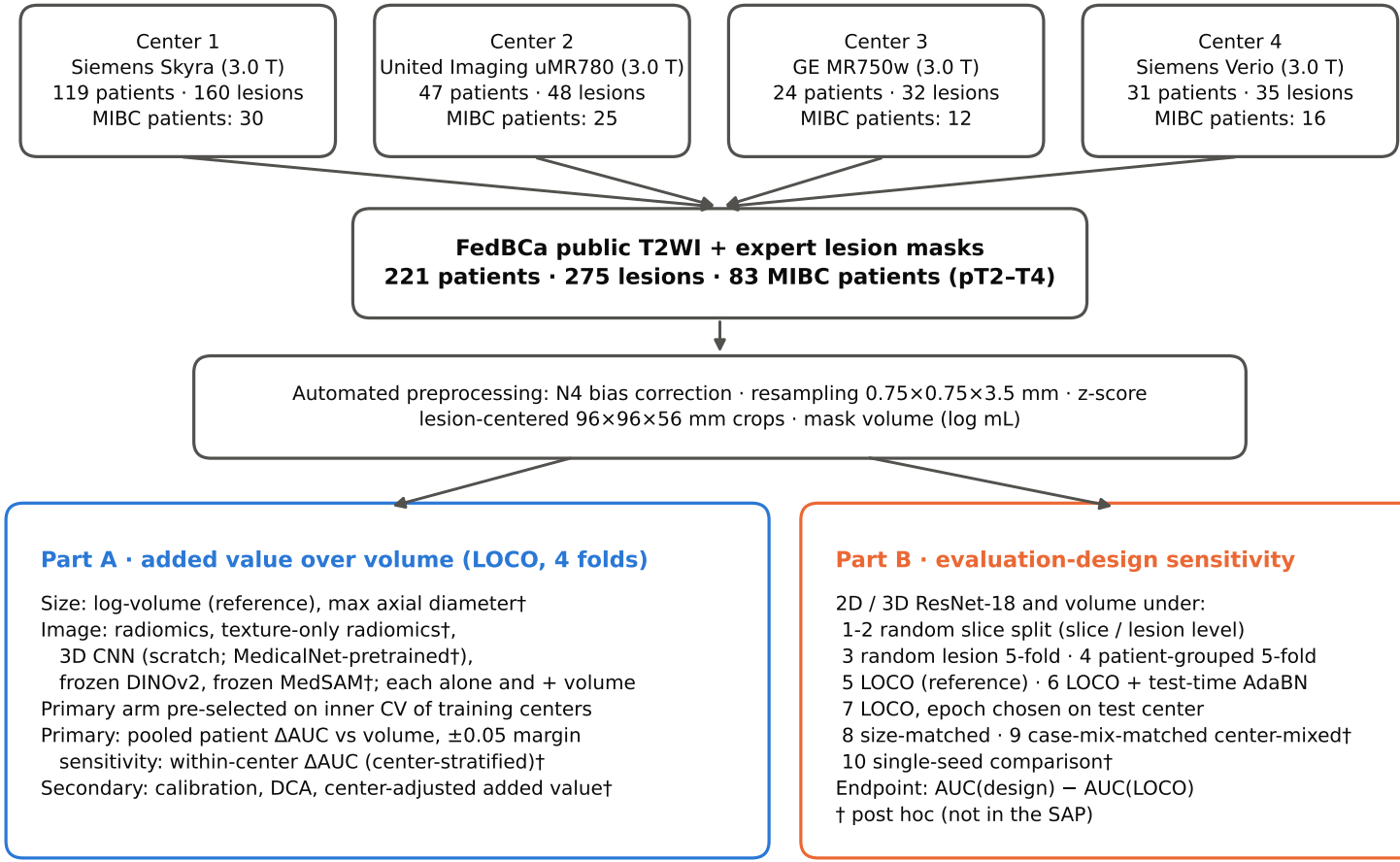
Center 1
Siemens Skyra (3.0 T)
119 patients · 160 lesions
MIBC patients: 30
Center 2
United Imaging uMR780 (3.0 T)
47 patients · 48 lesions
MIBC patients: 25
Center 3
GE MR750w (3.0 T)
24 patients · 32 lesions
MIBC patients: 12
Center 4
Siemens Verio (3.0 T)
31 patients · 35 lesions
MIBC patients: 16
FedBCa public T2WI + expert lesion masks
221 patients · 275 lesions · 83 MIBC patients (pT2–T4)
Automated preprocessing: N4 bias correction · resampling 0.75×0.75×3.5 mm · z-score
lesion-centered 96×96×56 mm crops · mask volume (log mL)
Part A · added value over volume (LOCO, 4 folds)
Size: log-volume (reference), max axial diameter†
Image: radiomics, texture-only radiomics†,
3D CNN (scratch; MedicalNet-pretrained†),
frozen DINOv2, frozen MedSAM†; each alone and + volume
Primary arm pre-selected on inner CV of training centers
Primary: pooled patient ΔAUC vs volume, ±0.05 margin
sensitivity: within-center ΔAUC (center-stratified)†
Secondary: calibration, DCA, center-adjusted added value†
Part B · evaluation-design sensitivity
2D / 3D ResNet-18 and volume under:
1-2 random slice split (slice / lesion level)
3 random lesion 5-fold · 4 patient-grouped 5-fold
5 LOCO (reference) · 6 LOCO + test-time AdaBN
7 LOCO, epoch chosen on test center
8 size-matched · 9 case-mix-matched center-mixed†
10 single-seed comparison†
Endpoint: AUC(design) − AUC(LOCO)
† post hoc (not in the SAP)